

# Revised Generation Interconnection System Impact Study Report

## For Queue Project Q12-0177

Scottsboro 100 kV  
January 01, 2014

### **1. Introduction**

This System Impact Study report documents the results of the analyses performed for interconnection request Q12-0177. The Interconnection Customer proposes a 201.0 MW solar photovoltaic facility interconnecting at the Scottsboro 100 kV substation in the Ohio area. The request was received on November 19, 2012. The study evaluated system intact and single contingency conditions using the regional planning case, including 137 higher-queued active requests and their assigned network upgrades.

### **2. Summary of Results**

The project was found to contribute (DFAX of 5% or greater) to 13 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$72,723,000.

### **3. Network Impacts**

#### **3.1 Generator Deliverability and Single Contingency Analysis**

1. Facility: Mormon Flat Dam – Scottsboro 100.0 kV  
Contingency: Scottsboro – Westrock Hodge La 100.0 kV Line  
Loading: 168.3% (Rating: 115 MVA)  
DFAX: 100.0%  
Pre-project loading: 6.8%
2. Facility: Scottsboro – Westrock Hodge A 100.0 kV  
Contingency: Scottsboro – Mormon Flat Dam 100.0 kV  
Loading: 159.7% (Rating: 121 MVA)  
DFAX: 100.0%  
Pre-project loading: 6.5%
3. Facility: Jackson Cntr Peaking – Camden Wire 100.0 kV  
Contingency: Scottsboro – Mormon Flat Dam Substation 100.0 kV Line  
Loading: 147.7% (Rating: 124 MVA)  
DFAX: 100.0%  
Pre-project loading: 14.6%
4. Facility: Westrock Hodge La – Jackson Cntr Peaking 100.0 kV  
Contingency: Mormon Flat Dam – Scottsboro 100.0 kV L  
Loading: 147.7% (Rating: 124 MVA)  
DFAX: 100.0%  
Pre-project loading: 14.6%
5. Facility: Eden Solar Llc Substation – Solon Diesel 100.0 kV  
Contingency: Camdn Wire – Adirondack 100.0 kV  
Loading: 117.6% (Rating: 139 MVA)  
DFAX: 100.0%  
Pre-project loading: 26.9%

6. Facility: Solon Diesel – Mormon Flat Dam Substation 100.0 kV  
Contingency: Camden Wire – Adirondack 100.0 kV  
Loading: 117.5% (Rating: 138 MVA)  
DFAX: 100.0%  
Pre-project loading: 27.7%
7. Facility: Eden Solar Llc Substation – Clapham 100.0 kV  
Contingency: Camden Wire – Adirondack 100.0 kV  
Loading: 112.9% (Rating: 135 MVA)  
DFAX: 100.0%  
Pre-project loading: 35.7%
8. Facility: Camden Wire – Adirondack 100.0 kV line  
Contingency: Eden Solar Llc – Solon Diesel 100.0 kV line  
Loading: 110.1% (Rating: 149 MVA)  
DFAX: 100.0%  
Pre-project loading: 25.1%
9. Facility: Piney Crk Project 500/345 kV XFMR  
Contingency: Hilltop – Piney Crk Project 500.0 kV  
Loading: 105.2% (Rating: 776 MVA)  
DFAX: 8.6%  
Pre-project loading: 103.0%
10. Facility: Talenenergy Martins Crk Llc Williamsport Substation 500/345 kV Transformer  
Contingency: Hilltop 500/345 kV Auto  
Loading: 101.4% (Rating: 1,035 MVA)  
DFAX: 6.2%  
Pre-project loading: 100.2%
11. Facility: Hilltop 500/345 kV XFMR  
Contingency: Talenenergy Martins Crk Llc Williamsport – Hilltop 500.0 kV Line  
Loading: 101.2% (Rating: 1,001 MVA)  
DFAX: 7.1%  
Pre-project loading: 99.8%
12. Facility: Carbon Center 500/345kV Transformer  
Contingency: Blackrock – Shawville Sta 345.0 kV  
Loading: 100.8% (Rating: 523 MVA)  
DFAX: 6.5%  
Pre-project loading: 98.3%
13. Facility: Little Chute – Talenenergy Martins Creek Llc Harrisburg 345.0 kV  
Contingency: Coronet – Ft Humbug 500.0 kV  
Loading: 100.6% (Rating: 1,230 MVA)  
DFAX: 6.6%  
Pre-project loading: 99.5%

### ***3.2 Multiple Facility Contingency Analysis***

No additional overloads were identified for multiple facility contingencies.

### ***3.3 Short Circuit and Stability***

Short circuit and stability analyses identified no additional network upgrades.

## ***4. Required Network Upgrades and Cost Estimates***

| Facility                                                             | Required Upgrade                            | Cost Estimate | Allocation to Project |
|----------------------------------------------------------------------|---------------------------------------------|---------------|-----------------------|
| Mormon Flat Dam – Scottsboro 100.0 kV                                | Rebuild line with higher-capacity conductor | \$10,053,000  | 100% (\$10,053,000)   |
| Scottsboro – Westrock Hodge A 100.0 kV                               | Rebuild line with higher-capacity conductor | \$2,291,000   | 100% (\$2,291,000)    |
| Camden Wire – Jackson Cntr Peaking 100.0 kV                          | Rebuild line with higher-capacity conductor | \$5,038,000   | 100% (\$5,038,000)    |
| Westrock Hodge La – Jackson Cntr Peaking 100.0 kV                    | Rebuild line with higher-capacity conductor | \$19,129,000  | 100% (\$19,129,000)   |
| Eden Solar Llc Substation – Solon Diesel 100.0 kV                    | Rebuild line with higher-capacity conductor | \$9,350,000   | 100% (\$9,350,000)    |
| Mormon Flat Dam – Solon Diesel 100.0 kV                              | Rebuild line with higher-capacity conductor | \$2,100,000   | 100% (\$2,100,000)    |
| Eden Solar Llc – Clapham 100.0 kV                                    | Rebuild line with higher-capacity conductor | \$12,911,000  | 100% (\$12,911,000)   |
| Camden Wire – Adirondack 100.0 kV line                               | Rebuild line with higher-capacity conductor | \$7,235,000   | 100% (\$7,235,000)    |
| Piney Crk Project 500/345 kV XFMR                                    | Replace transformer with higher-rated unit  | \$22,036,000  | 4% (\$797,000)        |
| Talenenergy Martins Creek Llc Williamsport Substation 500/345kV XFMR | Replace transformer with higher-rated unit  | \$34,043,000  | 4% (\$1,223,000)      |
| Hilltop 500/345 kV Auto                                              | Replace transformer with higher-rated unit  | \$24,135,000  | 4% (\$869,000)        |
| Carbon Ctr 500-345 kV TX                                             | Replace transformer with higher-rated unit  | \$39,361,000  | 4% (\$1,419,000)      |
| Little Chute – Talenenergy Martins Creek Llc Harrisburg 345.0 kV     | Reconductor line                            | \$8,584,000   | 4% (\$308,000)        |
|                                                                      | Total allocated to project                  |               | \$72,723,000          |

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.